

['FIST: Flow-inspired spatio-temporal target state modeling for visual tracking',

<https://www.sciencedirect.com/science/article/pii/S0031320326012082>,

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'Zhen Wan, Boxiong Sun, Tonghao Han, Yufei Zha, Peng Zhang',

'2',

'2026',

'12'],

['Switcher: Adaptive framework for unified and customized multi-modal object tracking',

<https://www.sciencedirect.com/science/article/pii/S0031320326009878>,

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'He Wang, Tianyang Xu, Zhangyong Tang, Xiao-Jun Wu, Josef Kittler',

'2',

'2026',

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['Occlusion-aware visual object tracking with explicit temporal state modeling and dual-memory mechanism',

<https://www.sciencedirect.com/science/article/pii/S0031320326011209>,

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'Zhipeng Chen, Linning Peng, Tianyu Wang, Cheng Zeng, Yijin Pan, Jun-Bo Wang',

'2',

'2026',

'12'],

['Causal Prompts for Open-Vocabulary Video Instance Segmentation',

<http://ieeexplore.ieee.org/document/11419811>,

'Open-vocabulary Video Instance Segmentation addresses the challenging task of detecting, segmenting, and tracking objects in videos, including categories not encountered during training. However, existing approaches often overlook rich temporal cues from preceding frames, limiting their ability to leverage causal context for robust open-world generalization. To bridge this gap, we propose CPOVIS, a novel framework that introduces causal prompts—dynamically propagated visual and taxonomy prompts from historical frames—to enhance temporal reasoning and semantic consistency. Built upon a Mask2Former architecture with a CLIP backbone, CPOVIS integrates three core innovations: 1) PromptCLIP, which aligns cross-modal embeddings while preserving open-vocabulary capabilities; 2) a Visual Prompt Injector that propagates object-level features to maintain spatial-temporal coherence; and 3) a Taxonomy Prompt Infuser that leverages hierarchical semantic relationships to stabilize unseen category recognition. Furthermore, we introduce a contrastive learning strategy to disentangle object representations across frames and adapt the Segment

Anything Model (SAM2) to boost open-vocabulary segmentation and tracking capacity in open-vocabulary video scenarios. Extensive experiments on seven challenging open- and closed-vocabulary video segmentation benchmarks demonstrate CPOVIS's state-of-the-art performance, outperforming existing methods by significant margins. Our findings highlight the critical role of causal prompt propagation in advancing video understanding in open-world scenarios.'

'Rongkun Zheng, Lu Qi, Xi Chen, Yi Wang, Kun Wang, Yu Qiao, Hengshuang Zhao',

'3',

'2026',

'6'],

['Unveiling the Power of Multi-Modal Template Update in RGBT Tracking',

<http://ieeexplore.ieee.org/document/11551799>'],

'Template update is essential for improving the adaptability of tracking algorithms to target appearance variations. While previous methods have leveraged the spatio-temporal complementarity of multi-modal templates for RGBT tracking, a comprehensive analysis of the template update mechanism remains underexplored. In this work, we propose a novel prototype-based framework that decomposes the multi-modal template update process from the perspective of prototype learning into four key components: multi-modal prototype, prototype integration, prototype evaluation, and prototype update algorithm. Our findings highlight that the multi-modal prototype is the most critical factor in enhancing tracking adaptability to appearance variations, leading to more robust target representations. While prototype integration is less crucial when the target representation is already robust, it still contributes to learning a more discriminative representation. Additionally, the accuracy of template updates is strongly influenced by prototype evaluation, which controls the accuracy of the update process. Finally, the prototype update algorithm, which determines when and how template updates occur, is key to maintaining tracking robustness. Building on these insights, we introduce the Multi-modal Prototype RGBT Tracker (MPTrack), which adapts dynamically to appearance variations through prototype learning. MPTrack combines a fixed template from the first frame with both modality-shared and modality-specific templates, forming a robust multi-modal prototype representation. It incorporates a prototype evaluation module that guides updates based on template reliability, and an adaptive update algorithm to manage templates effectively. Additionally, a prototype-guided cross-modal integration module enhances the discriminative power of multi-modal relation modeling. Experimental results on five challenging RGBT tracking benchmarks demonstrate that MPTrack consistently outperforms state-of-the-art methods, setting new performance records. The experimental data and source code will be made publicly available at: <https://github.com/mmic-lcl/Datasets-and-benchmark-code>.'

'Lei Liu, Chenglong Li, Andong Lu, Yabin Zhu, Shoufei Han, Xinye Cai, Changhe Li',

'4',

'2026',

'6'],

['Language Supervised Multi-Camera Multi-Object Tracking',

<http://ieeexplore.ieee.org/document/11554242>'],

'Recent multi-camera multi-object tracking (MCMOT) algorithms are primarily trained using per-detection identity annotations, which are complicated to obtain. In contrast, labeling a language description per-object is a more natural and human-friendly way. In this paper, we explore MCMOT in a language-supervised manner (LS-MCMOT) and propose a novel approach LaVST, which performs language-to-vision weakly-supervised learning based on reliable pseudo-labels generated via tracklet-level cross-modality matching. In addition, we design an ID-aware projection self-correction mechanism to correct inaccurate image-to-ground projection in a self-supervised manner. The models trained with our approach exhibit promising performance in LS-MCMOT. Surprisingly, they perform favorably against state-of-the-art identity-supervised methods, especially in cross-dataset evaluation (with an average gain by 20.0% in IDF1), underscoring the potential of language annotations in MCMOT. Codes and language annotations will be available here.'

'Kaige Mao, Xiaopeng Hong, Xiaopeng Fan, Wangmeng Zuo',

'4',

'2026',

'6'],

['Learning Occlusion-Dynamic Invariant Representations for Multi-Object Tracking',

<http://ieeexplore.ieee.org/document/11559212>'],

'Robust multi-object tracking (MOT) is hindered by the instability of appearance features under visual corruptions such as occlusion and motion blur. These perturbations introduce high-variance noise into feature trajectories, weakening temporal representations and causing identity switches. We address this challenge by learning more stable appearance representations under feature corruption. To this end, we propose the Causal Interaction Module (CIM), a causal architecture that follows a filter then reconstruct design for online tracking. A temporal filtering stage summarizes the historical feature trajectory into a stable anchor, and a contextual enhancement stage uses that anchor to refine frame-level features before association. Integrated into standard trackers, CIM improves association robustness while preserving the host tracking formulation. Experiments on multiple MOT benchmarks and corruption stress tests show consistent gains, especially on association-related metrics.'

'Muyu Li, Henan Hu, Deepak Kumar Jain, Ben Niu, Xudong Zhao',

'4',

'2026',

'6'],

['Adaptive Frequency-Based Constructive Wavelet Neural Network for Nonlinear Dynamic Systems',

<http://ieeexplore.ieee.org/document/11302811>'],

'Neural networks (NNs) have gained significant popularity for modeling complex, nonlinear systems due to their powerful approximation capabilities. However, designing an appropriate network structure and tuning parameters remains challenging, especially for nonlinear dynamic systems where offline training data are unavailable and poor approximations from badly tuned NNs can cause instability. This article presents a novel adaptive frequency-based constructive wavelet NN (AFBCWNN) for tracking reference trajectories for a class of unknown nonlinear dynamic systems. Using online measurements, the AFBCWNN integrates adaptive weight updating, adjustable network

structures, and rigorous stability analysis using Lyapunov techniques. Unlike conventional methods, the proposed AFBCWNN leverages frequency-domain analysis to estimate the energy distribution of the unknown nonlinear mapping from measured data. This frequency-based approach provides a uniform design guideline for network initialization, enabling the network to dynamically add wavelet bases when the desired accuracy is not achieved and prune nonenergy-active (low-energy) bases, reducing computational cost without compromising accuracy. Rigorous stability analysis establishes conditions for uniformly bounded trajectories, and simulation results confirm the AFBCWNN's superior performance in capturing complex, nonlinear dynamics compared to the existing adaptive methods.',

'Dunsheng Huang, Dong Shen, Lei Lu, Ying Tan',

'5',

'2026',

'6'],

['MtvTrack: Robust Visual Tracking via Modeling Time-Variant State of the Target',

<http://ieeexplore.ieee.org/document/11313329>'],

'Current single-object tracking algorithms depend on the information supplied by the template to identify and locate the object within the search area. However, environmental complexities and unknown factors can alter the object's state, causing mismatches in template information. The existing works using the template update mechanism (TUM) and multiple template feature fusion have the following problems: 1) TUM is affected by input superposition, making it hard to eliminate noise; 2) they suffer a temporal lag in their responsiveness to changes that occur in the object during the tracking process; 3) it is insufficient to rely solely on visual features within the search area of the current frame to improve the template; and 4) the prior knowledge regarding the input is not fully leveraged to learn the time-variant state of the object. We observe that in complex tracking scenarios, humans subconsciously analyze the evolutionary patterns of the object and its surroundings and integrate this information with the object's initial impression, thereby maintaining an awareness of the object's temporal state. Motivated by this, we propose a novel solution to the above problem, named MtvTrack, which can model the time-variant state of the object through the dynamic evolution pattern and static initial impression. Simultaneously, we propose a method for predicting the evolution pattern of scenes by utilizing past, present, and future (PPF) states. This approach effectively eliminates the information redundancy between consecutive frames and addresses the issue of delayed predictions of the target state in relation to changes within the search area. We establish a joint probability generative model and fully utilize prior knowledge to learn the time-variant state of the object. In addition, we develop a vector quantized-PPF (VQPPF) module for predicting the object's time-variant state. Experimental results on public benchmarks confirm the superior performance of our method. Source code is available at: <https://github.com/long-wa/MtvTrack-main>'],

'Jiaxu Zhao, Mingxing Jia, Lei Guo, Xingyu Han, Dapeng Niu',

'5',

'2026',

'6'],

['Reinforcement Learning-Based Optimal Formation Tracking for UAVs With Safety Constraints',

"We propose YOLOViT-DETECT, a hybrid model combining an enhanced YOLOv8 with a spatial attention module for fish detection and a ViT-B/16 transformer for species classification. The method achieves high accuracy in challenging underwater environments, outperforming standalone models. This approach enables real-time, reliable fish monitoring in aquaculture."

ABSTRACT

The ability to differentiate between different species of fish plays an essential role in aquaculture, as it helps to protect their populations and monitor their welfare. However, conventional learning methods are performance-limited, especially under complex submarine conditions such as similarity of fish body and background, low light levels and poor water quality. This paper proposes a new method titled YOLOViT-DETECT, which has practical significance for aquaculture monitoring by enabling accurate, automated tracking of fish species, supporting fish welfare, efficient farm management, and sustainable production. The approach is distinguished by its hybrid and attentional architecture which jointly optimizes the detection and classification of fish in complex underwater conditions. YOLOViT-DETECT uses an improved YOLOv8 model for the detection phase and a ViT-B/16 model for fish classification. To reinforce the model's focus on relevant areas of

attention, and Siou regression loss boost precision, especially for small objects. The model achieved a superior mAP of 94.9% on COCO, outperforming previous YOLO versions, and showed strong performance on PASCAL Visual Object Classes and LISA datasets.

ABSTRACT

For road safety, human factors and ergonomics play a vital role, influencing the object detection. To automate the object detection, several algorithms like R-CNN, SSD, R-FCN, and YOLO have been used for this task. While these models are effective for many object detection applications, detecting tiny objects on roads remains challenging due to low precision. Developing a lightweight and robust model capable of high-precision detection of tiny objects is essential. This study proposes a YOLO-O model, built on an enhanced YOLOv7 model, to detect tiny vehicles. A residual network with three convolutional layers and Squeeze and Excitation Network (SENet) replaces the Efficient Layer Aggregation Network (ELAN) in YOLOv7. Coordinate attention modules are added to improve recognition accuracy, and the Siou regression loss enhances bounding box prediction. Moreover, an 80×80 detection head improves the recognition of small vehicles. The model, tested on COCO, achieved superior performance with an mAP of 94.9%, outperforming YOLOv2, YOLOv3, and YOLOv5. Moreover, the proposed model performed significantly on cross-validation using the PASCAL Visual Object Classes and LISA datasets. This system demonstrates the method's practicality and robustness for tiny vehicle recognition, offering a foundation for developing automated object detection and tracking."

'Rabbia Mahum, Usama Irshad, Ahmed M. El-Sherbeeney, Mohammad Shehab, Seongkwan Mark Lee',

'7',

'2026',

'3'],

['Advancing Traffic Safety in Dense Urban Settings: A Robust Real-Time Wrong-Way Driving Detection System with Hough Transform and YOLOv5',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70322?af=R>,

'The study presents a real-time wrong-way driving detection system that integrates Hough Line Transform for road boundary extraction and YOLOv5 for vehicle detection, achieving 96% accuracy and a 0.05% false positive rate. Validated in Chattogram, Bangladesh, the system demonstrates a 60.05% detection rate across diverse urban traffic scenarios using a novel dataset of 100 videos. This innovative approach enhances traffic safety in dense urban settings by addressing the scarcity of context-specific data.

ABSTRACT

In Bangladesh, where a burgeoning population strains limited road networks, wrong-way driving (WWD) exacerbates traffic congestion and elevates accident risks, posing a critical public safety challenge. We present a pioneering real-time WWD detection system that integrates cutting-edge image processing and deep learning to transform traffic monitoring. Our methodology combines the Hough line transform for precise road boundary extraction with the state-of-the-art YOLOv5 algorithm for high-fidelity, real-time vehicle detection. Vehicle direction is inferred from single-frame orientation (distinguishing front/rear views) without multi-frame tracking, ensuring lightweight operation and scalable deployment. Tested on a novel, region-specific dataset from Chattogram, the system achieves an impressive 96% vehicle detection accuracy (mAP@0.5) during training and validation. Real-world evaluations on 20 diverse test videos yield an overall 60.6% recall for wrong-way vehicles, with a representative subset of 10 videos reaching 75% recall (and 41% precision). This research advances computer vision applications in traffic safety, providing an effective and deployable framework for reducing WWD incidents in high congestion urban regions and paving the way for enhanced intelligent transportation systems.'

To address the poor dynamic response and limited disturbance rejection of conventional dual-loop PI control in three-phase voltage source PWM rectifiers (VSRs), this paper proposes a composite control strategy featuring an improved sliding mode control (SMC) for the voltage outer-loop and a back-propagation neural network (BP-PID) controller for the current inner-loop. First, a mathematical model with feedforward decoupling is established in the synchronous rotating (dq) frame. For the DC-side voltage outer-loop, an improved SMC is designed by integrating an integral term into the sliding surface. This enhancement mitigates the slow response and steady-state error of conventional SMC under high-frequency disturbances and parameter variations. For the current inner-loop, a BP-PID adaptive controller is introduced to compensate for nonlinearities and improve current tracking precision, thereby reducing the impact of current fluctuations on the DC-link voltage. Finally, simulations are conducted in MATLAB/SIMULINK under step changes in voltage reference and load variations. The results verify that the proposed control strategy reduces the recovery time and

overshoot of the DC voltage, significantly improving both the dynamic response and the robustness against disturbances.

ABSTRACT

To address the poor dynamic response and limited disturbance rejection of conventional dual-loop PI control in three-phase voltage source PWM rectifiers (VSRs), this paper proposes a composite control strategy featuring an improved sliding mode control (SMC) for the voltage outer loop and a back-propagation neural network (BP-PID) controller for the current inner loop. First, a mathematical model with feedforward decoupling is established in the synchronous rotating (dq) frame. For the DC-side voltage outer loop, an improved SMC is designed by integrating an integral term into the sliding surface. This enhancement mitigates the slow response and steady-state error of conventional SMC under high-frequency disturbances and parameter variations. For the current inner loop, a BP-PID adaptive controller is introduced to compensate for nonlinearities and improve current tracking precision, thereby reducing the impact of current fluctuations on the DC-link voltage. Finally, simulations are conducted in MATLAB/SIMULINK under step changes in voltage reference and load variations. The results verify that the proposed control strategy reduces the recovery time and overshoot of the DC voltage, significantly improving both the dynamic response and the robustness against disturbances.'

'Xinyue Liang, Cheng Liu, Jiaqi Li',

'8',

'2026',

'6'],

['Robust Radar Tracking via Joint Adaptive EKF and Power Allocation Under Dynamic Clutter',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70601?af=R>,

ABSTRACT

In this letter, a joint adaptive extended Kalman filter (AEKF) and power allocation strategy is proposed for robust multitarget tracking in colocated MIMO radar systems under dynamic clutter. The unknown clutter intensity is incorporated into the measurement model and characterized by a truncated inverse gamma distribution. Using variational Bayesian inference, the proposed AEKF jointly estimates the target state and clutter intensity. Based on these estimates, a power allocation strategy is developed using the predicted conditional Cramér–Rao lower bound as a performance metric. Simulation results demonstrate that the proposed method achieves improved tracking performance in dynamic clutter environments.'

'Hao Sun, Yunhua Luo, Zhongjun Yu',

'8',

'2026',

'5'],

['Robust Sub-Filter Based APA With Parametric Welsch Function for Acoustic Echo Cancellation Under Impulsive Noise',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70595?af=R>,

'Acoustic echo cancellation (AEC) in actual communication systems is challenging due to highly correlated inputs, impulsive disturbances, and computational limitations. The traditional Affine Projection Algorithm (APA) is better than the Normalised Least Mean Square method, although it involves matrix inversion complexity and is susceptible to outliers (like impulsive noise). This paper presents a parametric Welsch-based APA with sub-filters to address these restrictions. The proposed

formulation exhibits superior performance without the necessity for matrix inversion calculations. This has been demonstrated through simulation under various settings.

ABSTRACT

The affine projection algorithm (APA) is widely used for acoustic echo cancellation, but faces challenges like slow convergence, high steady-state errors and sensitivity to impulsive noise. In this paper, a parametric Welsch APA has been introduced and integrated into the multiple sub-filter-based framework to address these issues for highly correlated input signals and impulsive noise. The employed parametric Welsch cost function robustly suppresses impulsive noise without any matrix inversion operation as in APA. On the other hand, a multiple sub-filter framework improves convergence and tracking performance. Through a theoretical stability and convergence analysis, the performance limits have been validated. Simulations performed using speech and synthetic signals with Bernoulli–Gaussian and impulsive noise demonstrate that the proposed approach achieves superior steady-state and convergence performance than existing APA variants.

Gagandeep Singh, Asutosh Kar, Vasundhara, Monorama Swain, Jesper Rindom Jensen, Mads Græsbøll Christensen,

'8',

'2026',

'5'],

['An Improved Robust Filter Algorithm for Manoeuvring Target Tracking With the Unknown and Time-Varying Noise',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70548?af=R>,

The unstable radar measurement noise in natural environments degrades tracking performance. This Letter introduces a noise-adaptive matrix to efficiently compute a corrected fading factor, enabling real-time compensation for noise variations. Superior performance is verified through comparison with existing methods.

ABSTRACT

In the natural environment, the unstable radar measurement noise can degrade the tracking performance of the manoeuvring target. In this Letter, the iterative formulation of the noise is simplified, and the noise-adaptive matrix is introduced to calculate the fading factor of the strong tracking algorithm so that the effect of measurement noise on the fading factor can be corrected in real-time. The superior performance of the proposed method is verified by comparison with three existing improved methods on a typical example.

Tianhao Liu, Xi Chen, Naichang Yuan,

'8',

'2026',

'3'],

['Compact Acoustic Signal Portraits for Resource-Efficient Mechanical Fault Detection',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70551?af=R>,

Dominant acoustic features are tracked to form compact signal portraits representing oscillatory modes. Faults are detected by portrait deviation, enabling low-complexity condition monitoring of moving machinery.

ABSTRACT

A resource-efficient semi-analytical method for mechanical fault detection via acoustic signal analysis is presented. A compact signal portrait is

controls the accuracy of the update process. Finally, the prototype update algorithm, which determines when and how template updates occur, is key to maintaining tracking robustness. Building on these insights, we introduce the Multi-modal Prototype RGBT Tracker (MPTrack), which adapts dynamically to appearance variations through prototype learning. MPTrack combines a fixed template from the first frame with both modality-shared and modality-specific templates, forming a robust multi-modal prototype representation. It incorporates a prototype evaluation module that guides updates based on template reliability, and an adaptive update algorithm to manage templates effectively. Additionally, a prototype-guided cross-modal integration module enhances the discriminative power of multi-modal relation modeling. Experimental results on five challenging RGBT tracking benchmarks demonstrate that MPTrack consistently outperforms state-of-the-art methods, setting new performance records. The experimental data and source code will be made publicly available at: <https://github.com/mmic-lcl/Datasets-and-benchmark-code>.

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[['Pixel-Level RGBT Fusion Tracking via Heterogeneous Multi-Expert Distillation and Decoupled Representation Learning',

<http://ieeexplore.ieee.org/document/11574556>,

'Pixel-level fusion is widely considered a lightweight yet limited strategy in RGB-Thermal (RGBT) tracking due to its shallow representational capacity. However, its actual limitations and potential remain largely unexplored. We systematically analyze fusion location, modality alignment, and tracking performance, revealing that despite lower modality gaps than feature-level fusion, pixel-level fusion lacks task-relevant discrimination, restricting its effectiveness. In this paper, we propose the Task-driven Pixel-level Fusion tracker (TPF), which preserves the efficiency of early fusion while enhancing discriminative capacity. Central to TPF is a lightweight pixel fusion adapter that ensures real-time image fusion with only 14.3KB extra parameters over the baseline at inference. To enhance its limited representational capacity, we propose a task-driven progressive learning framework consisting of two key stages. First, a heterogeneous multi-expert distillation scheme adaptively transfers image fusion knowledge from diverse models under tracking-guided evaluation, mitigating the generalization limitations of single-teacher distillation across varied tracking scenarios. Second, to overcome limited task discrimination caused by sparse, target-focused tracking supervision, we propose a decoupled representation learning strategy that offers dense, complementary guidance to improve target-background separation and fusion quality. A nearest-neighbor dynamic template update further enhances robustness to appearance changes. Extensive experiments on four RGBT tracking benchmarks show that TPF achieves competitive accuracy and speed, outperforming both feature-level and existing pixel-level fusion methods, offering new insights into efficient RGBT tracking.'

'Andong Lu, Yuanzhi Guo, Kunpeng Wang, Chenglong Li, Jin Tang, Bin Luo',

'4',

'2026',

'71],

['MMCATrack: Multi-Modal Channel Attention Tracker',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/cvi2.70060?af=R>,

'We proposes a multi-modal channel attention tracking algorithm wherein a multi-modal channel attention block is incorporated for the purpose of enhancing the representation ability of the key features within the multi-modal features.

ABSTRACT

Most existing Transformer-based visual object tracking methods rely exclusively on the feature map from the last encoder layer for object prediction, thereby overlooking the rich information contained in shallow and intermediate layer feature maps. This limitation reduces the representational capacity of the model. Moreover, current multi-modal tracking frameworks typically construct multi-modal features through simple concatenation, which fails to adequately account for the differential contributions of individual modalities to the final prediction task. As a result, these approaches exhibit an insufficient ability to express key features within the multi-modal representation. To address the aforementioned issues, this paper proposes a multi-modal channel attention tracking algorithm, where a multi-modal

[['Cell as Point: One-stage framework for efficient cell tracking',

<https://www.sciencedirect.com/science/article/pii/S0031320326013415>,

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'Yaxuan Song, Jianan Fan, Heng Huang, Mei Chen, Weidong Cai',

'2',

'2026',

'12'],

['Pragmatic Communication in Multi-Agent Collaborative Perception',

<http://ieeexplore.ieee.org/document/11478377>,

'Collaborative perception allows each agent to enhance its perceptual abilities by exchanging messages with others. It inherently results in a trade-off between perception ability and communication costs. Previous works transmit complete full-frame high-dimensional feature maps among agents, resulting in substantial communication costs. To promote communication efficiency, we propose only transmitting the information needed for the collaborator's downstream task. This pragmatic communication strategy focuses on three key aspects: i) pragmatic message selection, which selects task-critical parts from the complete data, resulting in spatially and temporally sparse feature vectors; ii) pragmatic message representation, which achieves pragmatic approximation of high-dimensional feature vectors with a task-adaptive dictionary, enabling communicating with integer indices; iii) pragmatic collaborator selection, which identifies beneficial collaborators, pruning unnecessary communication links. Following this strategy, we first formulate a mathematical optimization framework for the perception-communication trade-off and then propose PragComm, a multi-agent collaborative perception system with two key components: i) single-agent detection and tracking and ii) pragmatic collaboration. The proposed PragComm promotes pragmatic communication and adapts to a wide range of communication conditions. We evaluate PragComm for both collaborative 3D object detection and tracking tasks in both real-world, V2V4Real, and simulation datasets, OPV2V and V2X-SIM2.0. PragComm consistently outperforms previous methods with more than 32.7 K× lower communication volume on OPV2V.'

'Yue Hu, Xianghe Pang, Xiaoqi Qin, Yonina C. Eldar, Siheng Chen, Ping Zhang, Wenjun Zhang',

'3',

'2026',

'7'],

['Motion2VecSets: Non-Rigid Shape Reconstruction and Tracking With 4D Latent Set Diffusion',

<http://ieeexplore.ieee.org/document/11475204>,

'We introduce Motion2VecSets, a 4D diffusion model for dynamic surface mesh generation from various ambiguous observations, including a sequence of RGB images, sparse and partial point clouds, and low-resolution voxel grids. While recent methods using neural field representations have shown success in modeling non-rigid objects, conventional feed-forward architectures struggle with noisy, partial, or sparse observations due to their deterministic nature. To address the inherent one-to-many mapping problem, we introduce a diffusion model that explicitly learns the shape and motion distribution of non-rigid objects through an iterative denoising process of compressed latent representations. The diffusion-based priors provide more plausible and diverse reconstructions under ambiguous conditions. Instead of relying on global latent codes, we represent 4D dynamics using

latent sets. This novel 4D representation captures local shape and deformation patterns, leading to more accurate non-linear motion capture and significantly improving generalization capacity to unseen motions and identities. For temporally coherent tracking, we jointly denoise latent sets across frames and enable cross-frame information exchange. To reduce computational cost, we design an interleaved spatial-temporal attention block that alternately aggregates deformation latents along spatial and temporal dimensions. Extensive experiments on datasets of humans, animals, and articulated objects demonstrate that Motion2VecSets outperforms prior methods in reconstructing and tracking non-rigid deformations from various imperfect observations.'

'Jiapeng Tang, Wei Cao, Biao Zhang, Chang Luo, Yaoyao Liu, Matthias Nießner',

'3',

'2026',

'7'],

['Probing Effective and Efficient Category-Level Articulated Object Pose Perception',

<http://ieeexplore.ieee.org/document/11480457>'],

'Category-level articulated object pose perception—encompassing both static pose estimation and dynamic pose tracking—is critical for embodied AI systems interacting with complex environments. Due to the inherent complexity and diverse motion structures of articulated objects, existing methods often exhibit limitations in adequately modeling kinematic constraints, handling self-occlusions, and meeting optimization requirements. Building upon EfficientCAPER (Yu et al., 2024), this work introduces CAPER++, a unified framework addressing these limitations through three key innovations: first, a joint-centric hierarchical model decomposes objects into a root part and constrained parts linked by joints, explicitly embedding kinematic constraints for geometrically consistent pose recovery. Second, an $SE(3)$ manifold formulation leverages Lie algebra in the tangent space for singularity-free rotation representation and stable optimization, replacing error-prone direct regression. Third, for tracking, a proxy canonicalization strategy reformulates pose updates as $SE(3)$ increment predictions relative to keyframes, enhanced by a dynamic keyframe mechanism to suppress drift. Extensive experiments on synthetic (ArtImage, PM-Videos), semi-synthetic (ReArtMix, ReArt-Videos), and real-world (RobotArm, RobotArm-Videos) benchmarks demonstrate state-of-the-art accuracy and robustness. CAPER++ achieves real-time inference (50 FPS) without post-processing, significantly advancing category-level articulated perception for real-world applications.'

'Li Zhang, Xianhui Meng, Liu Liu, Haonan Jiang, Jianan Wang, Rujing Wang, Cewu Lu, Jun Liu, Hong Zhang',

'3',

'2026',

'7'],

['Pixel-Level RGBT Fusion Tracking via Heterogeneous Multi-Expert Distillation and Decoupled Representation Learning',

<http://ieeexplore.ieee.org/document/11574556>'],

'Pixel-level fusion is widely considered a lightweight yet limited strategy in RGB-Thermal (RGBT) tracking due to its shallow representational capacity. However, its actual limitations and potential remain largely unexplored. We systematically analyze fusion location, modality alignment, and

tracking performance, revealing that despite lower modality gaps than feature-level fusion, pixel-level fusion lacks task-relevant discrimination, restricting its effectiveness. In this paper, we propose the Task-driven Pixel-level Fusion tracker (TPF), which preserves the efficiency of early fusion while enhancing discriminative capacity. Central to TPF is a lightweight pixel fusion adapter that ensures real-time image fusion with only 14.3KB extra parameters over the baseline at inference. To enhance its limited representational capacity, we propose a task-driven progressive learning framework consisting of two key stages. First, a heterogeneous multi-expert distillation scheme adaptively transfers image fusion knowledge from diverse models under tracking-guided evaluation, mitigating the generalization limitations of single-teacher distillation across varied tracking scenarios. Second, to overcome limited task discrimination caused by sparse, target-focused tracking supervision, we propose a decoupled representation learning strategy that offers dense, complementary guidance to improve target-background separation and fusion quality. A nearest-neighbor dynamic template update further enhances robustness to appearance changes. Extensive experiments on four RGBT tracking benchmarks show that TPF achieves competitive accuracy and speed, outperforming both feature-level and existing pixel-level fusion methods, offering new insights into efficient RGBT tracking.'

'Andong Lu, Yuanzhi Guo, Kunpeng Wang, Chenglong Li, Jin Tang, Bin Luo',

'4',

'2026',

'7'],

['Symmetric Entropy-Constrained Video Coding for Machines',

<http://ieeexplore.ieee.org/document/11578184>'],

'As video transmission increasingly serves machine vision systems (MVS) instead of human vision systems (HVS), video coding for machines (VCM) has become a critical research topic. Existing VCM methods often bind codecs to specific downstream models, requiring retraining or supervised data, thus limiting generalization in multi-task scenarios. Recently, unified VCM frameworks have employed visual backbones (VB) and visual foundation models (VFM) to support multiple video understanding tasks with a single codec. They mainly utilize VB/VFM to maintain semantic consistency or suppress non-semantic information, but seldom explore how to directly link video coding with understanding under VB/VFM guidance. Hence, we propose a Symmetric Entropy-Constrained Video Coding framework for Machines (SEC-VCM). It establishes a symmetric alignment between the video codec and VB, allowing the codec to leverage VB's representation capabilities to preserve semantics and discard MVS-irrelevant information. Specifically, a bi-directional entropy-constraint (BiEC) mechanism ensures symmetry between the process of video decoding and VB encoding by suppressing conditional entropy. This helps the codec to explicitly handle semantic information beneficial to MVS while squeezing useless information. Furthermore, a semantic-pixel dual-path fusion (SPDF) module injects pixel-level priors into the final reconstruction. Through semantic-pixel fusion, it suppresses artifacts harmful to MVS and improves machine-oriented reconstruction quality. Experimental results on classical video understanding tasks and MLLM-based tasks show state-of-the-art (SOTA) rate-task performance. It achieves significant bitrate savings over H.266/VVC reference software VTM on video instance segmentation (37.4%), video object segmentation (29.8%), object detection (46.2%), multiple object tracking (44.9%), and MLLM-based video grounding (97.6%). The code is at <https://github.com/Ws-Syx/SEC-VCM>,'

'Yuxiao Sun, Meiqin Liu, Chao Yao, Qi Tang, Jian Jin, Weisi Lin, Frederic Dufaux, Yao Zhao',

function to develop a rock bolt point detection network, which effectively enhances the detection and localisation accuracy. Finally, a two-stage serial fitting method is proposed to solve the linear equation for each row and complete the measurement. Simulation experiments demonstrate that the developed point detection network achieves notable gains in precision, recall, and positioning error. The precision is 87.53%, the recall is 93.03%, and the positioning error is 11.33mm, all ranking among the top performers. Industrial experiments demonstrate that the method is well-suited for on-site applications.'

'You Lei, Yang Xiaoyu, Luo Chuan, Li Tao, Zhang Haifeng, Liu Yanchi',

'7',

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'5'],

['YOLOViT-DETECT: A Hybrid Approach for Real-Time Fish Species Detection and Classification in Aquaculture',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70373?af=R>,

"We propose YOLOViT-DETECT, a hybrid model combining an enhanced YOLOv8 with a spatial attention module for fish detection and a ViT-B/16 transformer for species classification. The method achieves high accuracy in challenging underwater environments, outperforming standalone models. This approach enables real-time, reliable fish monitoring in aquaculture.

ABSTRACT

The ability to differentiate between different species of fish plays an essential role in aquaculture, as it helps to protect their populations and monitor their welfare. However, conventional learning methods are performance-limited, especially under complex submarine conditions such as similarity of fish body and background, low light levels and poor water quality. This paper proposes a new method titled YOLOViT-DETECT, which has practical significance for aquaculture monitoring by enabling accurate, automated tracking of fish species, supporting fish welfare, efficient farm management, and sustainable production. The approach is distinguished by its hybrid and attentional architecture which jointly optimizes the detection and classification of fish in complex underwater conditions. YOLOViT-DETECT uses an improved YOLOv8 model for the detection phase and a ViT-B/16 model for fish classification. To reinforce the model's focus on relevant areas of the image, even in the presence of a complex background, we have added a spatial attention module (SAM) to YOLOv8. Detection results from YOLOv8 Improved and classification results from ViT were merged into a single output. Experiments have shown that YOLOViT-DETECT is a robust method whose results exceed those achieved with YOLOv8 and ViT-B/16 separately. The precision rate reached 0.9033 in the detection phase and 0.9524 in the classification phase."

'Mahdi Hamzaoui, Mohamed Ould-Elhassen Aoueileyine, Imen Filali, Ridha Bouallegue',

'7',

'2026',

'4'],

['PhantomVeil: Physical Camouflage for Multi-Object Tracking Deep Neural Networks',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70353?af=R>,

'PhantomVeil, a physical camouflage attack method for multi-object tracking, systematically exploits the structural vulnerabilities in the association between detection and tracking modules, thereby

constructing a physically realizable adversarial generation

framework.

ABSTRACT

Multi-object tracking (MOT) models have achieved great success in wide ranging applications, such as video surveillance and autonomous driving, especially for those deep neural networks (DNNs) based MOT, which obtained the state-of-the-art detection performance. Unfortunately, existing research has unveiled that DNNs are vulnerable to carefully crafted adversarial perturbations, leading to catastrophic consequences. In order to explore the adversarial vulnerability of MOT, we propose PhantomVeil, a novel physical camouflage for DNN based MOT. PhantomVeil significantly differs from previous work in four aspects: (1) effectiveness—by undermining the data association between the detection and tracking stages of MOT models, our method reliably ensures a high probability of attack success; (2) generality—by targeting the common core mechanisms of MOT models, we design an efficient loss optimization strategy that enables generalized attacks across different models; (3) transferability—Gaussian smoothing and color gamut constraints are introduced, it reduces the adversarial attenuation when transferring from the digital domain to the physical domain and improves the effectiveness of target transfer in the physical domain by restricting drastic pixel value fluctuations and ensuring printable color values; (4) practicability—extensive experiments conducted on two public datasets and one self-built dataset demonstrate that PhantomVeil outperforms four baseline methods on three multi-object trackers (with digital-domain attack performance reaching $\times 1.02 \sim 6.07$ that of the baselines and physical-domain attack performance reaching $\times 2.19 \sim 6.07$ that of the baselines), verifying its advantages in practicality, effectiveness, and generalization.

Jinyin Chen, Huanran Li, Ye Han',

'7',

'2026',

'4'],

['YOLO-O: An Improved YOLO-Based Framework for Vehicle Detection',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70337?af=R>,

"This study proposes a YOLO-O model, based on YOLOv7, for improved detection of tiny vehicles on roads. Enhancements like residual networks, Squeeze and Excitation Network (SENet), coordinate attention, and SIoU regression loss boost precision, especially for small objects. The model achieved a superior mAP of 94.9% on COCO, outperforming previous YOLO versions, and showed strong performance on PASCAL Visual Object Classes and LISA datasets.

ABSTRACT

For road safety, human factors and ergonomics play a vital role, influencing the object detection. To automate the object detection, several algorithms like R-CNN, SSD, R-FCN, and YOLO have been used for this task. While these models are effective for many object detection applications, detecting tiny objects on roads remains challenging due to low precision. Developing a lightweight and robust model capable of high-precision detection of tiny objects is essential. This study proposes a YOLO-O model, built on an enhanced YOLOv7 model, to detect tiny vehicles. A residual network with three convolutional layers and Squeeze and Excitation Network (SENet) replaces the Efficient Layer Aggregation Network (ELAN) in YOLOv7. Coordinate attention modules are added to improve recognition accuracy, and the SIoU regression loss enhances bounding box prediction. Moreover, an 80×80 detection head improves the recognition of small vehicles. The model, tested on COCO, achieved superior performance with an mAP of 94.9%, outperforming YOLOV2, YOLOV3, and YOLOV5. Moreover, the proposed model performed significantly on cross-validation using the PASCAL Visual Object Classes and LISA datasets. This system demonstrates the method's practicality and robustness

control strategy featuring an improved sliding mode control (SMC) for the voltage outer-loop and a back-propagation neural network (BP-PID) controller for the current inner-loop. First, a mathematical model with feedforward decoupling is established in the synchronous rotating (dq) frame. For the DC-side voltage outer-loop, an improved SMC is designed by integrating an integral term into the sliding surface. This enhancement mitigates the slow response and steady-state error of conventional SMC under high-frequency disturbances and parameter variations. For the current inner-loop, a BP-PID adaptive controller is introduced to compensate for nonlinearities and improve current tracking precision, thereby reducing the impact of current fluctuations on the DC-link voltage. Finally, simulations are conducted in MATLAB/SIMULINK under step changes in voltage reference and load variations. The results verify that the proposed control strategy reduces the recovery time and overshoot of the DC voltage, significantly improving both the dynamic response and the robustness against disturbances.

ABSTRACT
To address the poor dynamic response and limited disturbance rejection of conventional dual-loop PI control in three-phase voltage source PWM rectifiers (VSRs), this paper proposes a composite control strategy featuring an improved sliding mode control (SMC) for the voltage outer loop and a back-propagation neural network (BP-PID) controller for the current inner loop. First, a mathematical model with feedforward decoupling is established in the synchronous rotating (dq) frame. For the DC-side voltage outer loop, an improved SMC is designed by integrating an integral term into the sliding surface. This enhancement mitigates the slow response and steady-state error of conventional SMC under high-frequency disturbances and parameter variations. For the current inner loop, a BP-PID adaptive controller is introduced to compensate for nonlinearities and improve current tracking precision, thereby reducing the impact of current fluctuations on the DC-link voltage. Finally, simulations are conducted in MATLAB/SIMULINK under step changes in voltage reference and load variations. The results verify that the proposed control strategy reduces the recovery time and overshoot of the DC voltage, significantly improving both the dynamic response and the robustness against disturbances.'

'Xinyue Liang, Cheng Liu, Jiaqi Li',

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'2026',

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['Robust Radar Tracking via Joint Adaptive EKF and Power Allocation Under Dynamic Clutter',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70601?af=R>,

ABSTRACT
In this letter, a joint adaptive extended Kalman filter (AEKF) and power allocation strategy is proposed for robust multitarget tracking in colocated MIMO radar systems under dynamic clutter. The unknown clutter intensity is incorporated into the measurement model and characterized by a truncated inverse gamma distribution. Using variational Bayesian inference, the proposed AEKF jointly estimates the target state and clutter intensity. Based on these estimates, a power allocation strategy is developed using the predicted conditional Cramér–Rao lower bound as a performance metric. Simulation results demonstrate that the proposed method achieves improved tracking performance in dynamic clutter environments.'

'Hao Sun, Yunhua Luo, Zhongjun Yu',

'8',

'2026',

'5'],

['Robust Sub-Filter Based APA With Parametric Welsch Function for Acoustic Echo Cancellation Under Impulsive Noise',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70595?af=R>,

'Acoustic echo cancellation (AEC) in actual communication systems is challenging due to highly correlated inputs, impulsive disturbances, and computational limitations. The traditional Affine Projection Algorithm (APA) is better than the Normalised Least Mean Square method, although it involves matrix inversion complexity and is susceptible to outliers (like impulsive noise). This paper presents a parametric Welsch-based APA with sub-filters to address these restrictions. The proposed formulation exhibits superior performance without the necessity for matrix inversion calculations. This has been demonstrated through simulation under various settings.'

ABSTRACT

The affine projection algorithm (APA) is widely used for acoustic echo cancellation, but faces challenges like slow convergence, high steady-state errors and sensitivity to impulsive noise. In this paper, a parametric Welsch APA has been introduced and integrated into the multiple sub-filter-based framework to address these issues for highly correlated input signals and impulsive noise. The employed parametric Welsch cost function robustly suppresses impulsive noise without any matrix inversion operation as in APA. On the other hand, a multiple sub-filter framework improves convergence and tracking performance. Through a theoretical stability and convergence analysis, the performance limits have been validated. Simulations performed using speech and synthetic signals with Bernoulli–Gaussian and impulsive noise demonstrate that the proposed approach achieves superior steady-state and convergence performance than existing APA variants.'

'Gagandeep Singh, Asutosh Kar, Vasundhara, Monorama Swain, Jesper Rindom Jensen, Mads Græsbøll Christensen',

'8',

'2026',

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['An Improved Robust Filter Algorithm for Manoeuvring Target Tracking With the Unknown and Time-Varying Noise',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70548?af=R>,

'The unstable radar measurement noise in natural environments degrades tracking performance. This Letter introduces a noise-adaptive matrix to efficiently compute a corrected fading factor, enabling real-time compensation for noise variations. Superior performance is verified through comparison with existing methods.'

ABSTRACT

In the natural environment, the unstable radar measurement noise can degrade the tracking performance of the manoeuvring target. In this Letter, the iterative formulation of the noise is simplified, and the noise-adaptive matrix is introduced to calculate the fading factor of the strong tracking algorithm so that the effect of measurement noise on the fading factor can be corrected in real-time. The superior performance of the proposed method is verified by comparison with three existing improved methods on a typical example.'

'Tianhao Liu, Xi Chen, Naichang Yuan',

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<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70551?af=R>,

'Dominant acoustic features are tracked to form compact signal portraits representing oscillatory modes. Faults are detected by portrait deviation, enabling low-complexity condition monitoring of moving machinery.'

ABSTRACT A resource-efficient semi-analytical method for mechanical fault detection via acoustic signal analysis is presented. A compact signal portrait is constructed by tracking dominant salient-frequency peaks over time, without a priori assumptions on peak locations. Peak feature statistics are aggregated into a 3D matrix that encodes nominal oscillatory modes. Fault detection is achieved by measuring the deviation between current and healthy portraits using a single decision threshold. Experiments on internal combustion engine recordings demonstrate improved separability and lower computational cost compared to empirical mode decomposition, confirming suitability for resource-constrained monitoring.'

'Oleksandr Volkov, Oleksandr Bushma, Artem Sieriebriakov, Volodymyr Taranukha, Yaroslav Linder',

'8',

'2026',

'2'],

['A Novel Quantised Image Sensing for Machine Vision',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70515?af=R>,

This paper shows that the image information quantised in the range of 1 to 4 bits can be utilised for object detection and tracking without any performance degradation. Thus, it is expected that the proposed technique holds promise for use in machine vision, not only maintaining the accuracy of detection and tracking, but even enabling more optimised information transfer between the image sensor and the computing system.

ABSTRACT

This paper proposes a quantisation-driven data-compression framework for machine vision sensors by combining nonlinear quantisation and bit-shift masking. The proposed approach compresses pixel data to an effective 1–4 bits per pixel while preserving edge-salient information required for object detection and feature tracking. Experimental results show that the compressed/quantised images maintain detection and tracking performance without degradation, indicating the proposed framework can reduce sensor-to-processor bandwidth and optimise required bit resolution.'

'Paul K. J. Park',

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'2026',

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